

Team Les Misérables

CSE 4408: System Analysis and Design
Lab 7: 30/07/2025

Members

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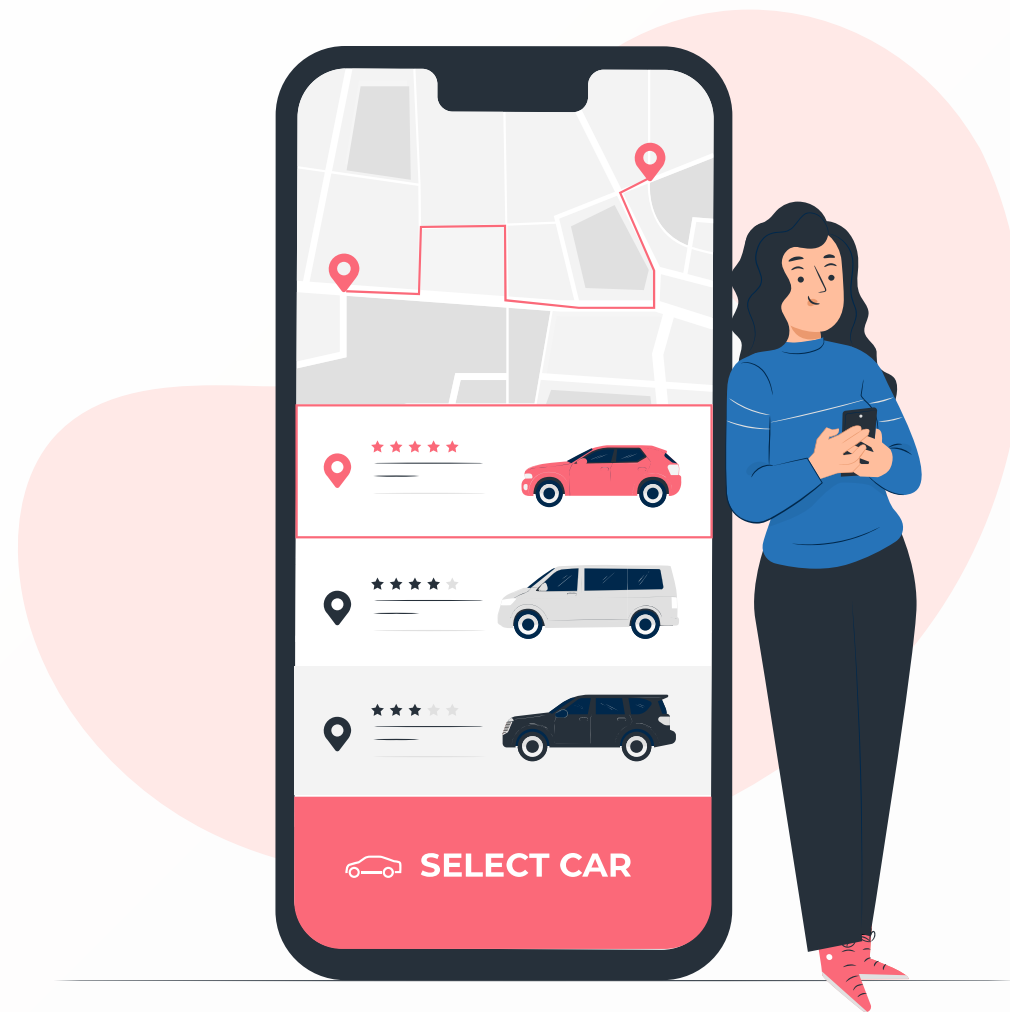


BashayJabo

Because the best part of IUT is going home



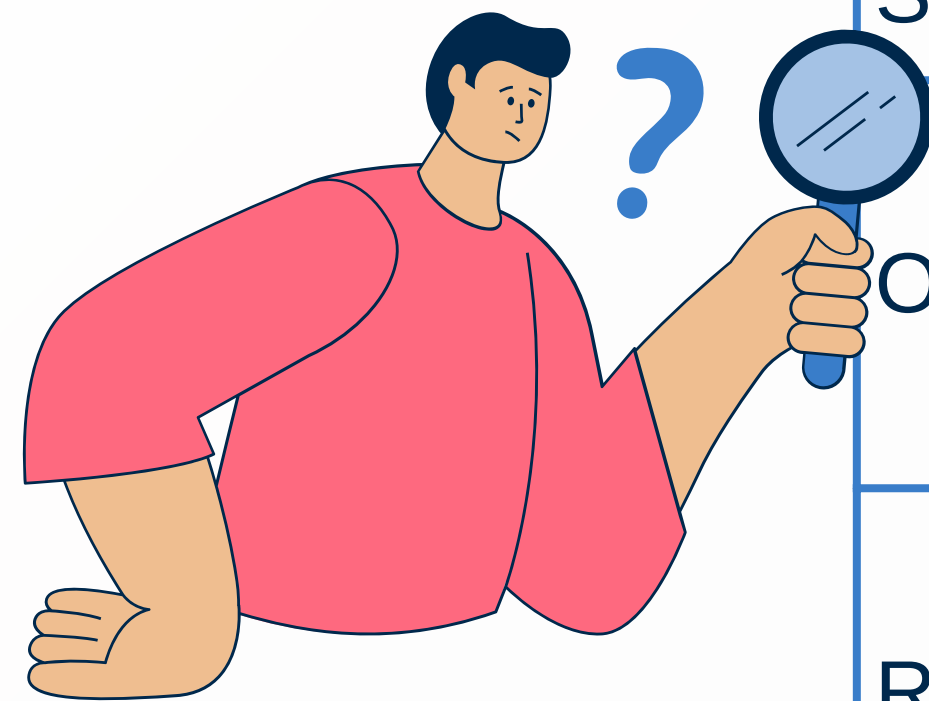
Overview



This system is being developed by an independent startup founded by us. Though not officially a part of the university, the startup is dedicated to serving the IUT student community.

Its core mission is to ease everyday student challenges through innovative digital solutions.

Stakeholder Needs



Stakeholder	Needs
Frequently Commuting Students	Reliable, low-cost rides with minimal coordination hassle.
Occasional Travelers	Flexibility and visibility into available rides without long-term commitments.
Regular Organisers	Frustrated by manually messaging peers, fare splitting confusion, and no-shows.

Identified Process Flows

Current ride coordination is mostly manual through WhatsApp / Facebook group or individual chats.

No centralized way to track or confirm ride status, often causing confusion or missed rides.



1. Riders create informal posts / text messages.
 - List transport type, location, other preferences.
2. Wait for responses from peers.
3. Discuss fares and pickup timing with confirmed participants.
4. Follow up with updates or changes.
 - Repeat steps if plans fall through.

Derived Features

- Ride creation and browsing/joining
- Preference filters (gender, department, schedule)
- Fare split calculation
- Online payment via Bkash/Nagad
- Ride confirmation system
- Status updates/notifications
- Login via university email
- Integration with Uber/Shohoz





MVP Feature Selection

1. Ride Posting:

User Story: As a **ride provider**, I want to post details of my ride (time, location, seats), so that others can view and join if it matches their schedule.

Priority: Must

Justification: Foundational functionality. Without this, there are no rides in the system.

2. Ride Browsing and Joining:

User Story: As a **ride seeker**, I want to browse and join available rides, so that I can commute easily without personally asking multiple people.

Priority: Must

Justification: Core user action. Ride-sharing cannot happen if students can't join posted rides.



MVP Feature Selection

3. User Authentication & Verification:

User Story: As a **ride seeker**, I want to verify that other users are IUT students, so that I can feel safer when traveling with someone new.

Priority: Must

Justification: Trust and safety are essential, prevents misuse. Will keep the app IUT-exclusive.

4. Ride Preference Matching (Optional Filters):

User Story: As a **commuter**, I want to filter rides by gender or department, so that I can travel with people I'm more comfortable with.

Priority: Should

Justification: Improves personalization, but system works without it initially.



MVP Feature Selection

5. Location Selection via Map Integration:

User Story: As a ride provider, I want to pin pickup and drop-off locations on a map, so others can easily find the meeting point without confusion.

Priority: Could

Justification: Important for clarity but can be replaced initially with drop-down location fields.

6. Ride Notifications (Real-time updates):

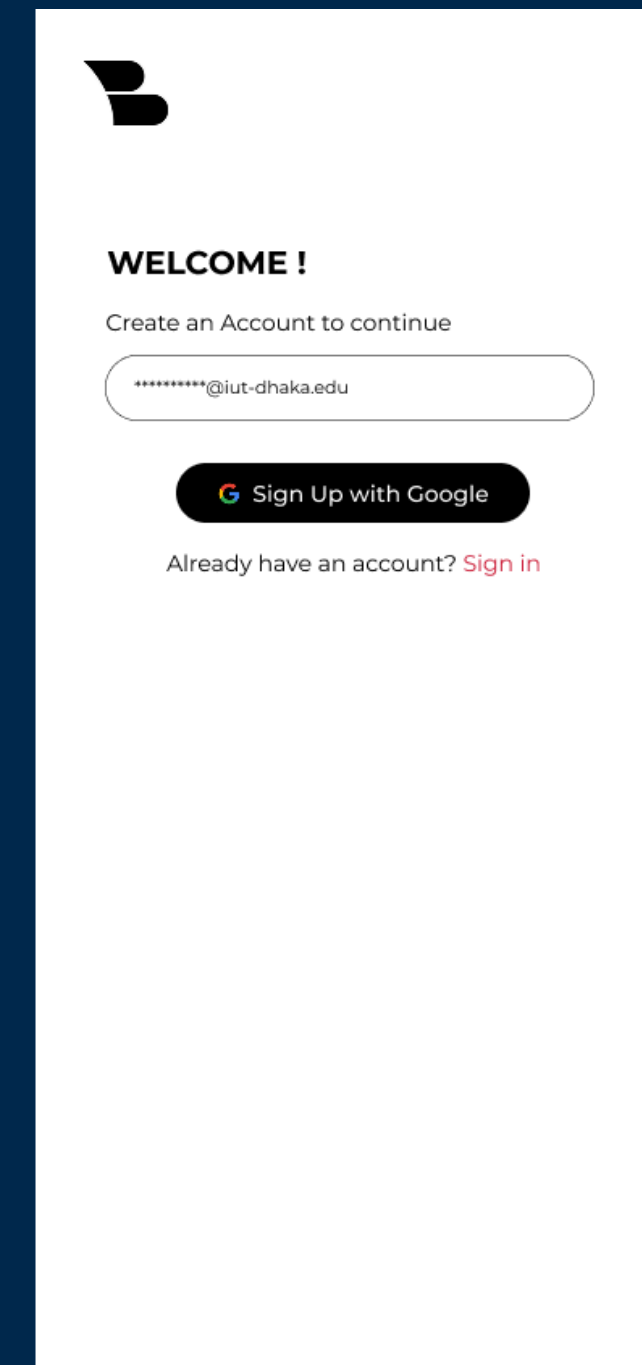
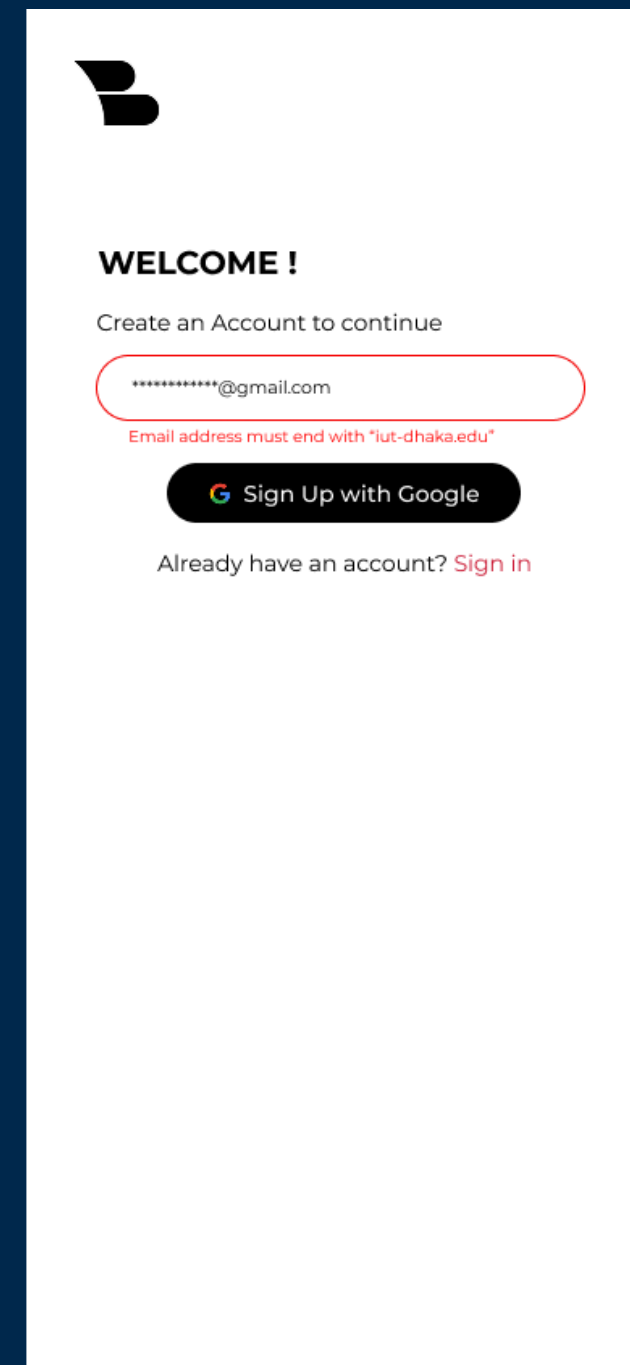
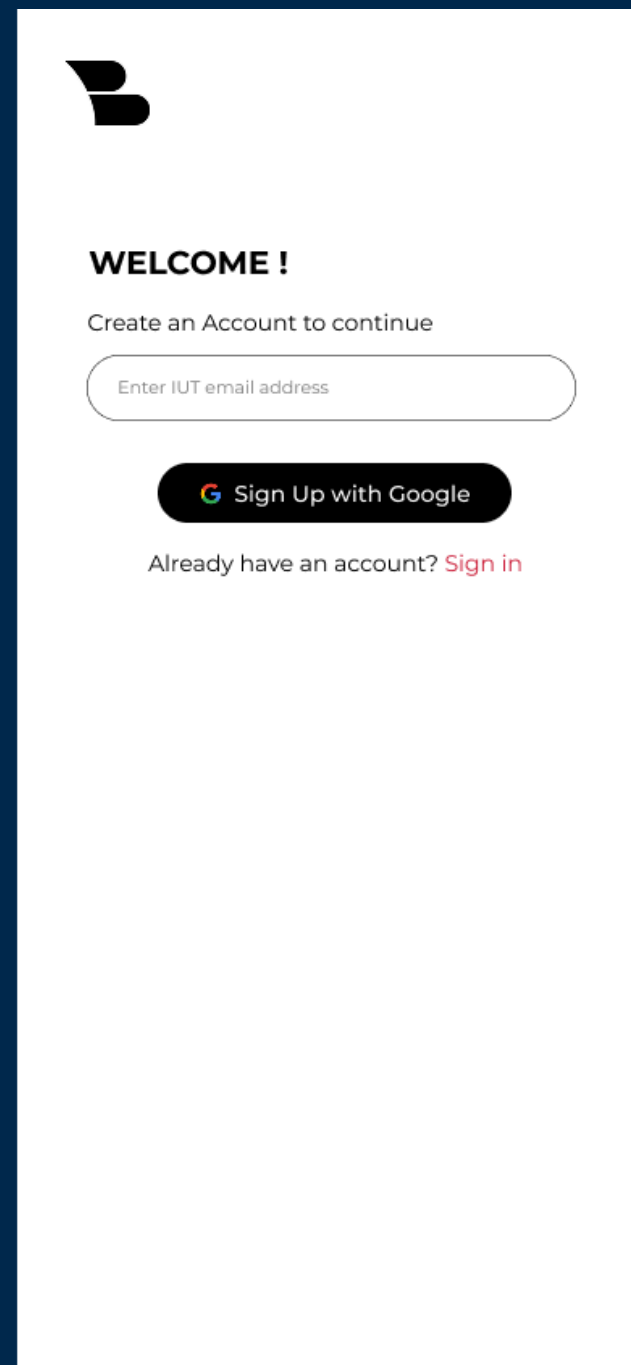
User Story: As a ride participant, I want to receive real-time updates when rides are modified or filled, so that I don't miss important changes or arrive too late.

Priority: Could

Justification: Enhances usability but not essential to launch, improves user experience.

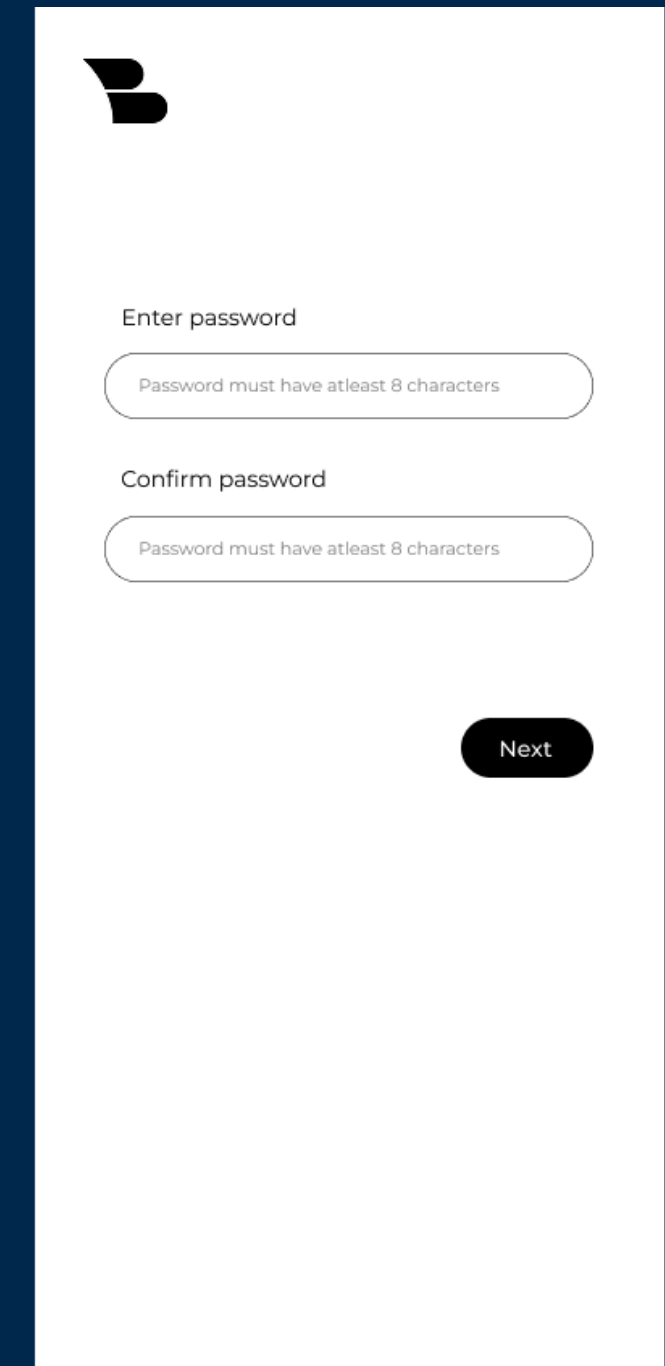
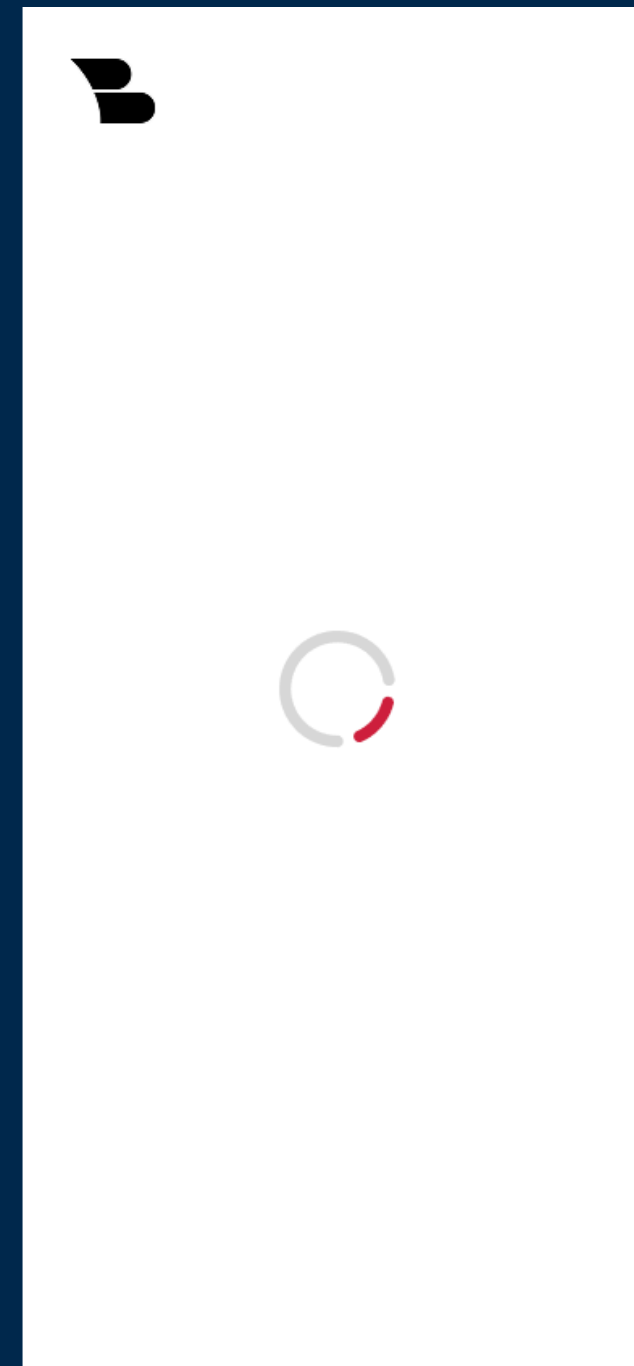
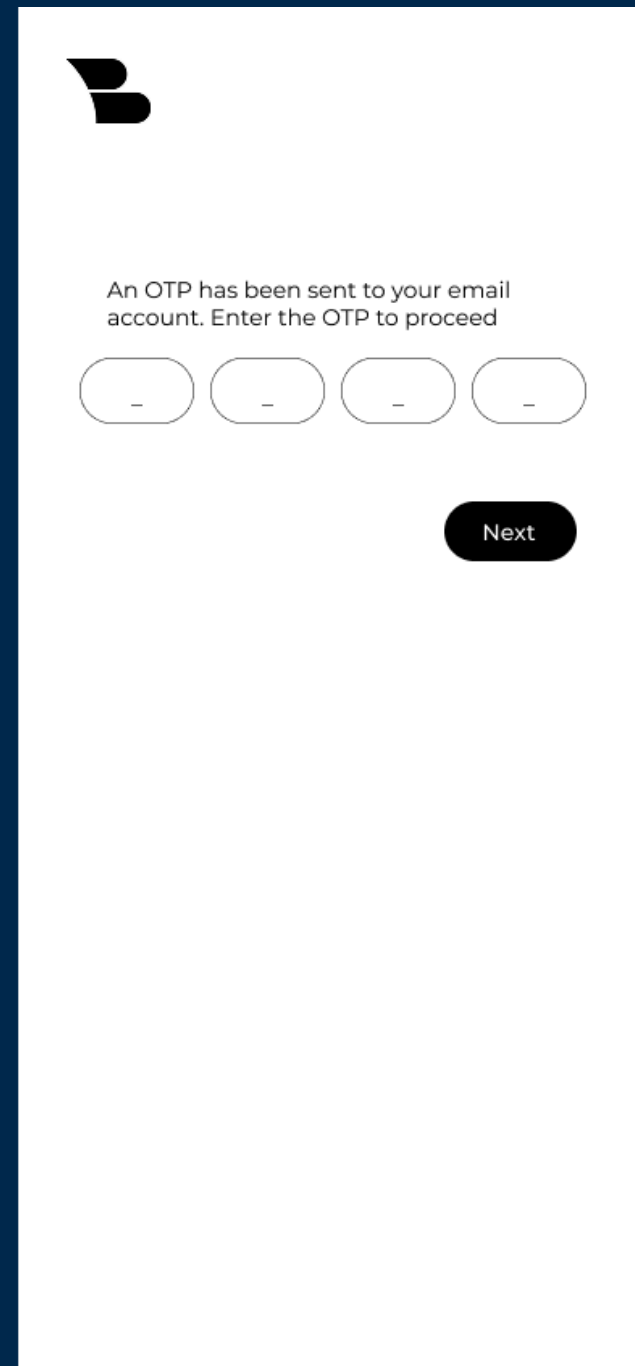
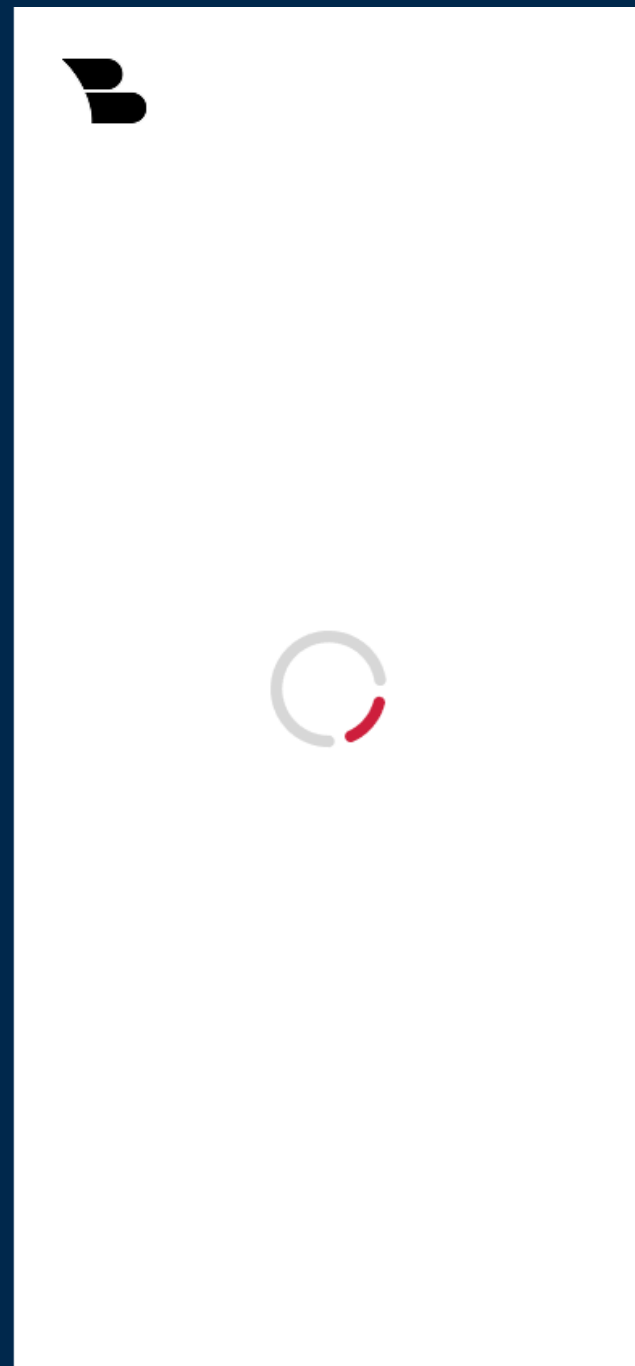
Prototyping

1. User Authentication & Verification



Prototyping

1. User Authentication & Verification



Prototyping

1. User Authentication & Verification





Enter User Information

Username

Contact number

Student ID

Department

Gender

☐ Male ☐ Female

Address

Next



Add a profile picture



Skip for now

Next

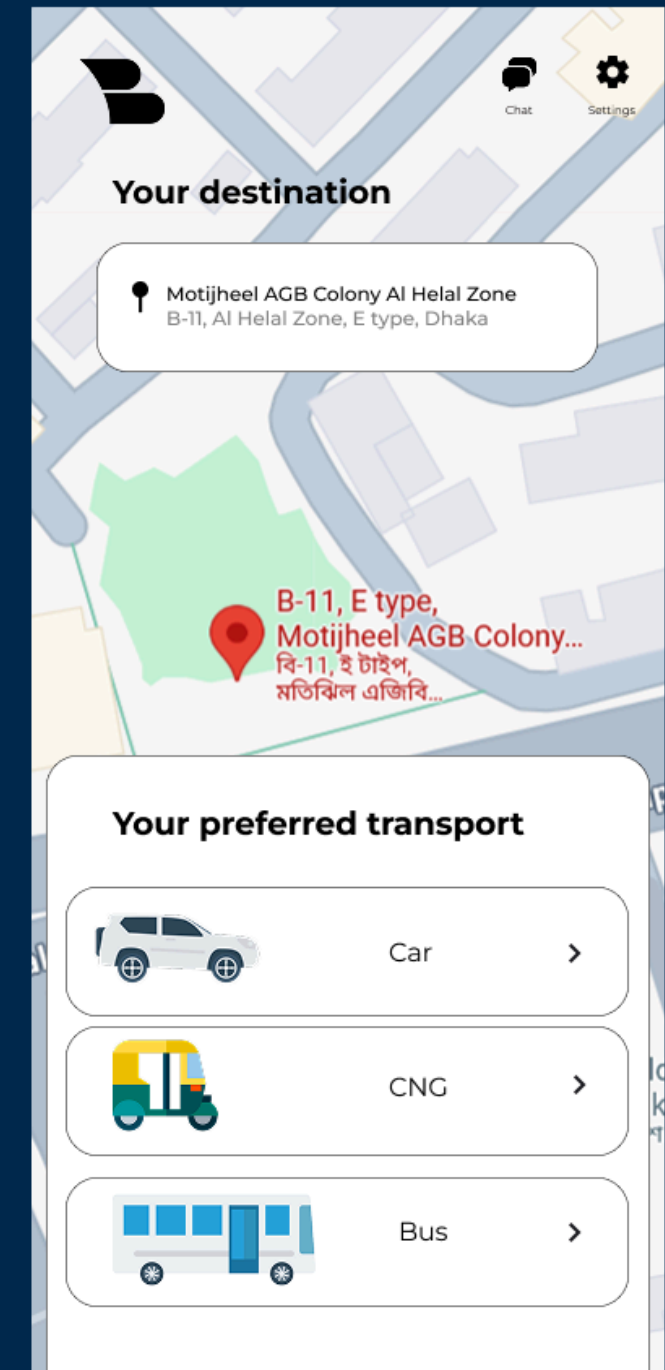
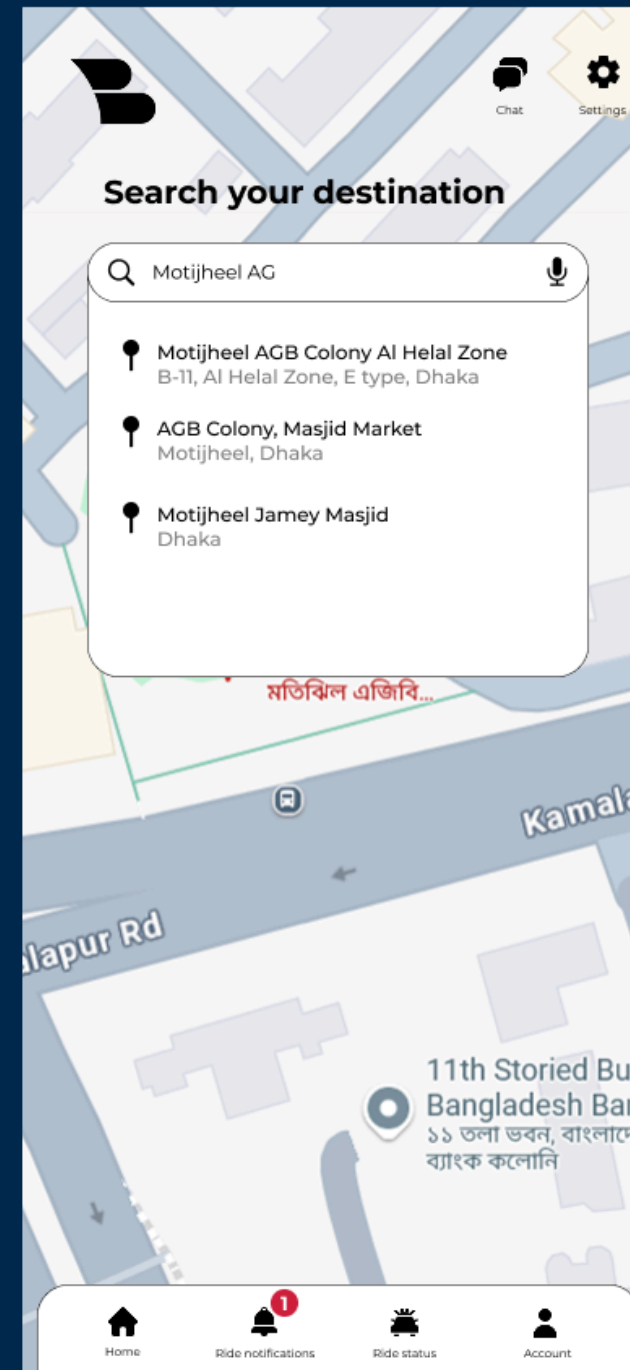
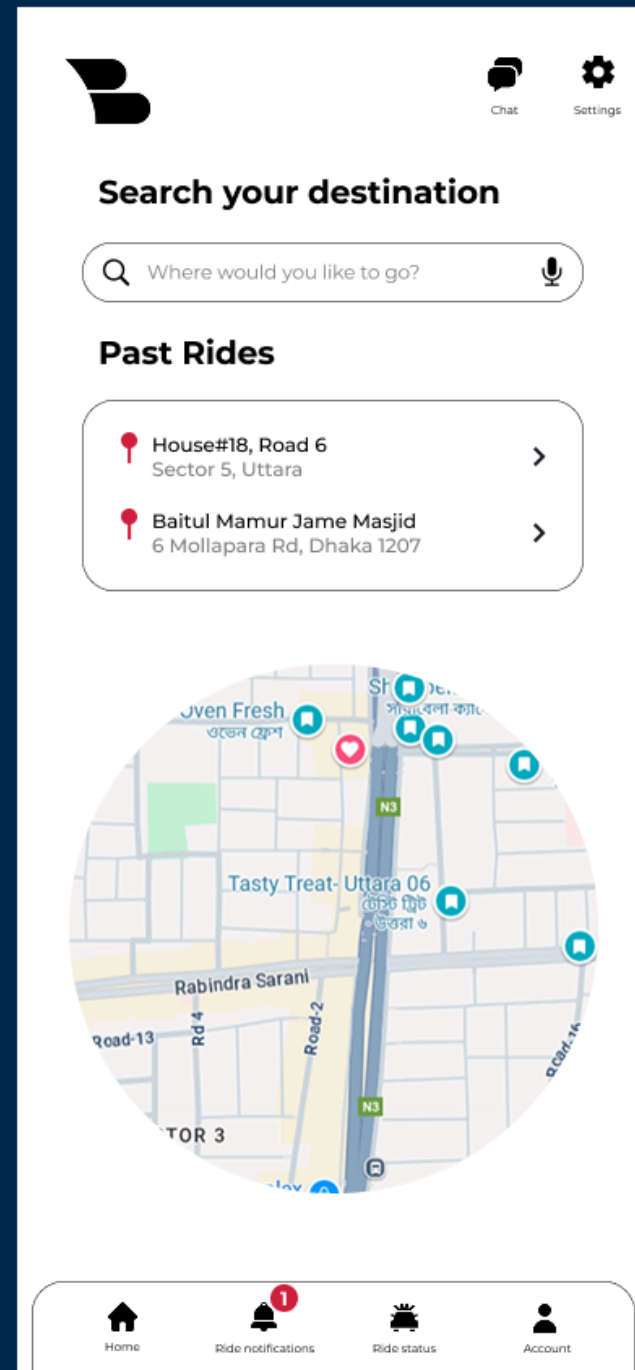
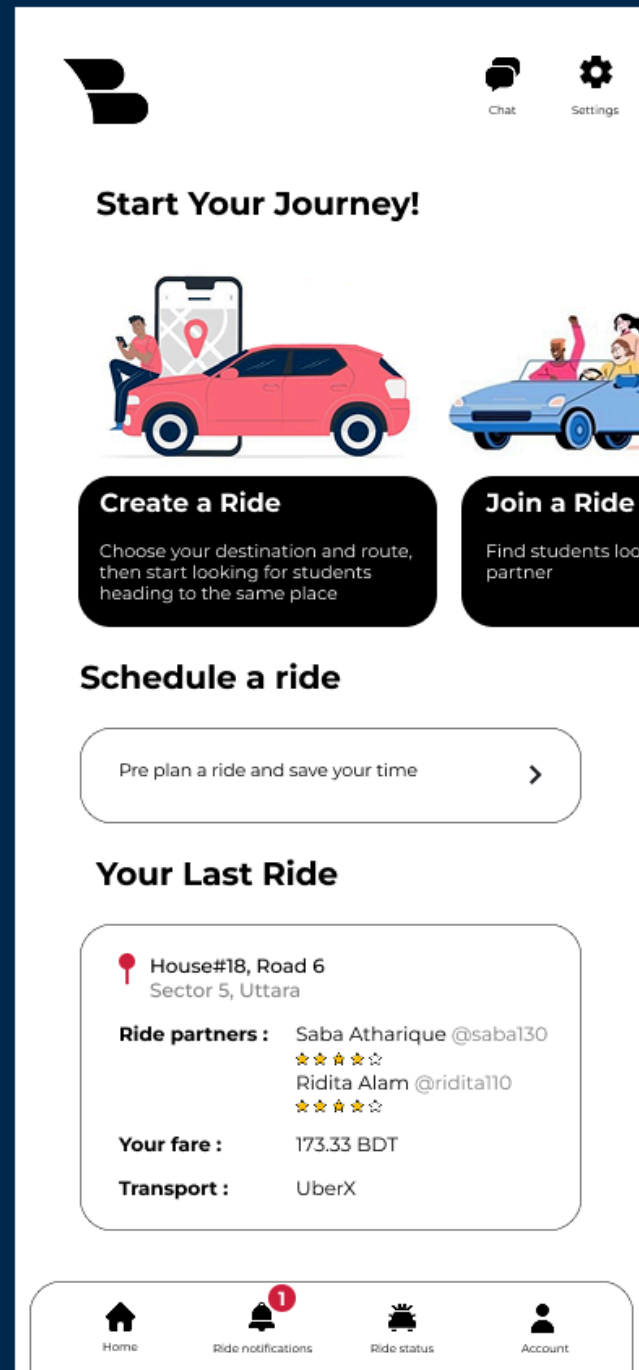


You're Set Up!

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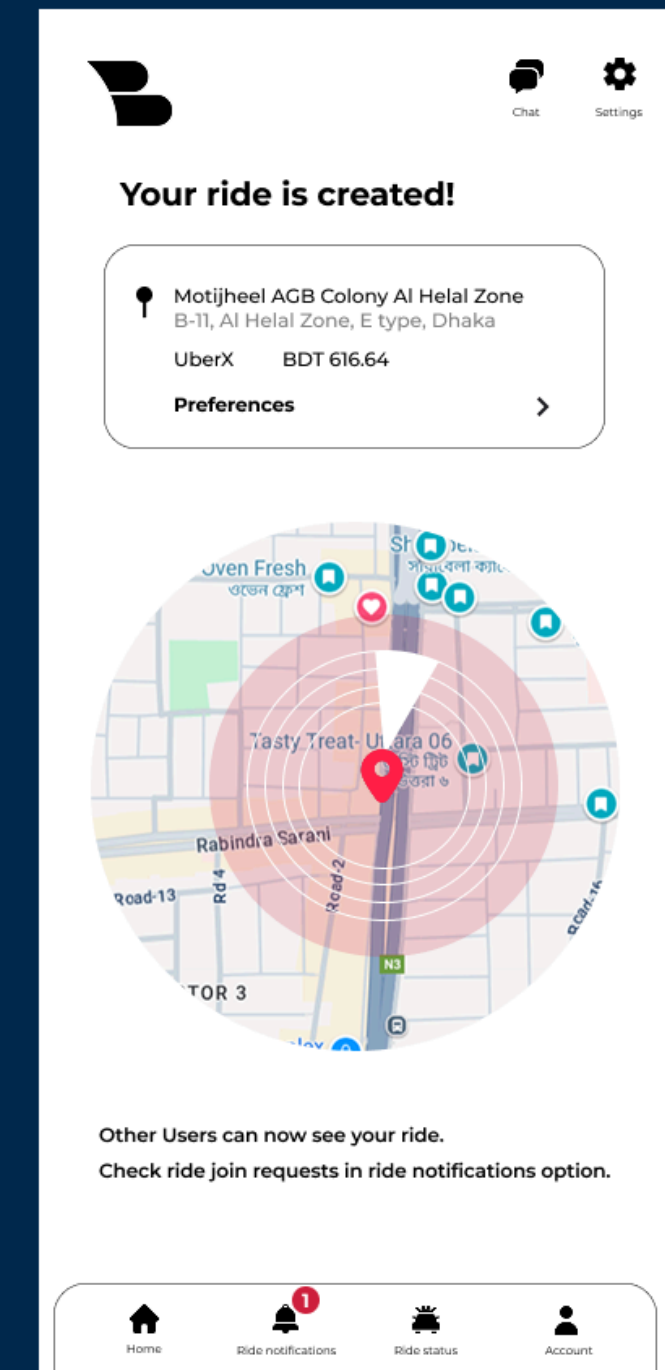
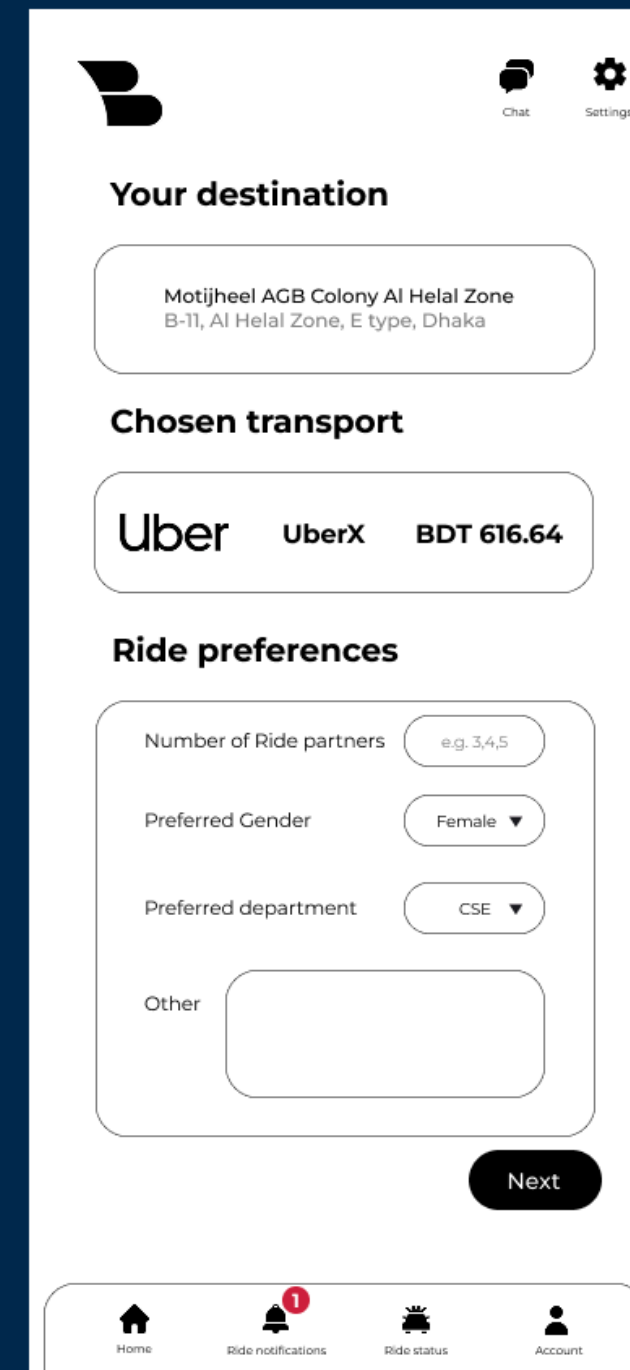
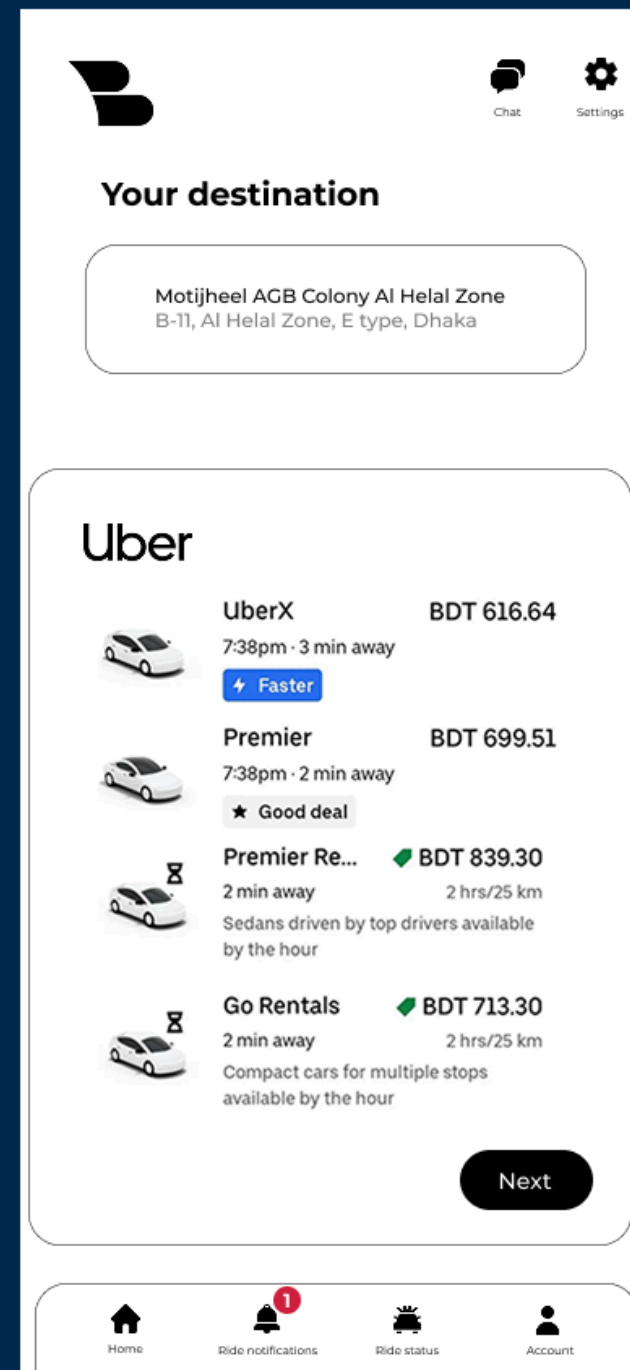
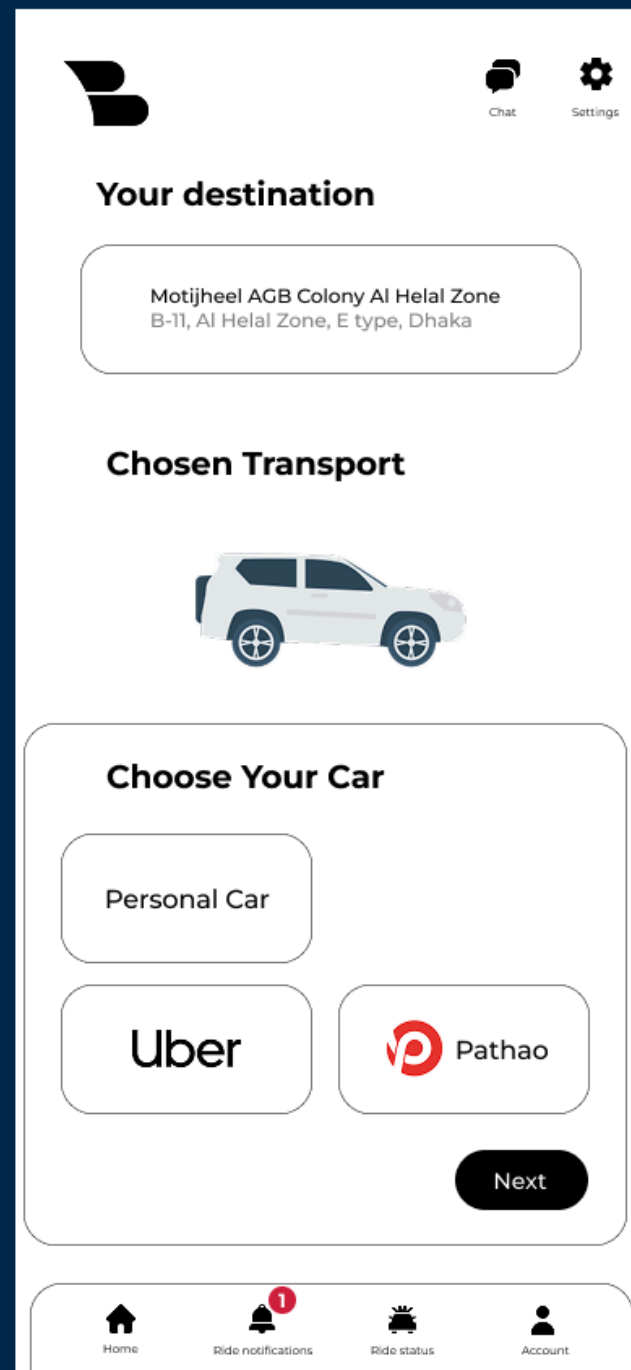
Prototyping

2. Ride Posting



Prototyping

2. Ride Posting



Scrum Roles and Team Structure

Role	Member	Justification
Product Owner	Ridita	• Led stakeholder interviews • Understanding of student pain points • Prioritize features accordingly
Scrum Master	Saba	• Highly organized, kept track of tasks during previous labs • Facilitates communication
Development Team Member	Mounota	• Experienced with UI design and mobile app development • Mockups, prototyping, & integration



MoSCoW & Product Backlog

Feature	Priority	Justification	Size
Ride Posting	Must	Enables core ride contribution	M
Ride Browsing & Joining	Must	Enables core ride participation	M
User Authentication	Must	Ensures safety and trust	S
Ride Preference & Matching	Should	Improves user comfort and satisfaction	L
Map-Based Location Selection	Could	Enhances clarity and reduces miscommunication	M
Real-time Ride Notifications	Could	Improves coordination and UX	M

First 2-week Sprint

User Story	Reason for Selection	Story Points
User Authentication & Verification	Essential for account-based access and safety.	3
Ride Posting	Core function that enables users to contribute rides.	5
Ride Browsing & Joining	Paired with posting, this completes the core ride-sharing loop.	5

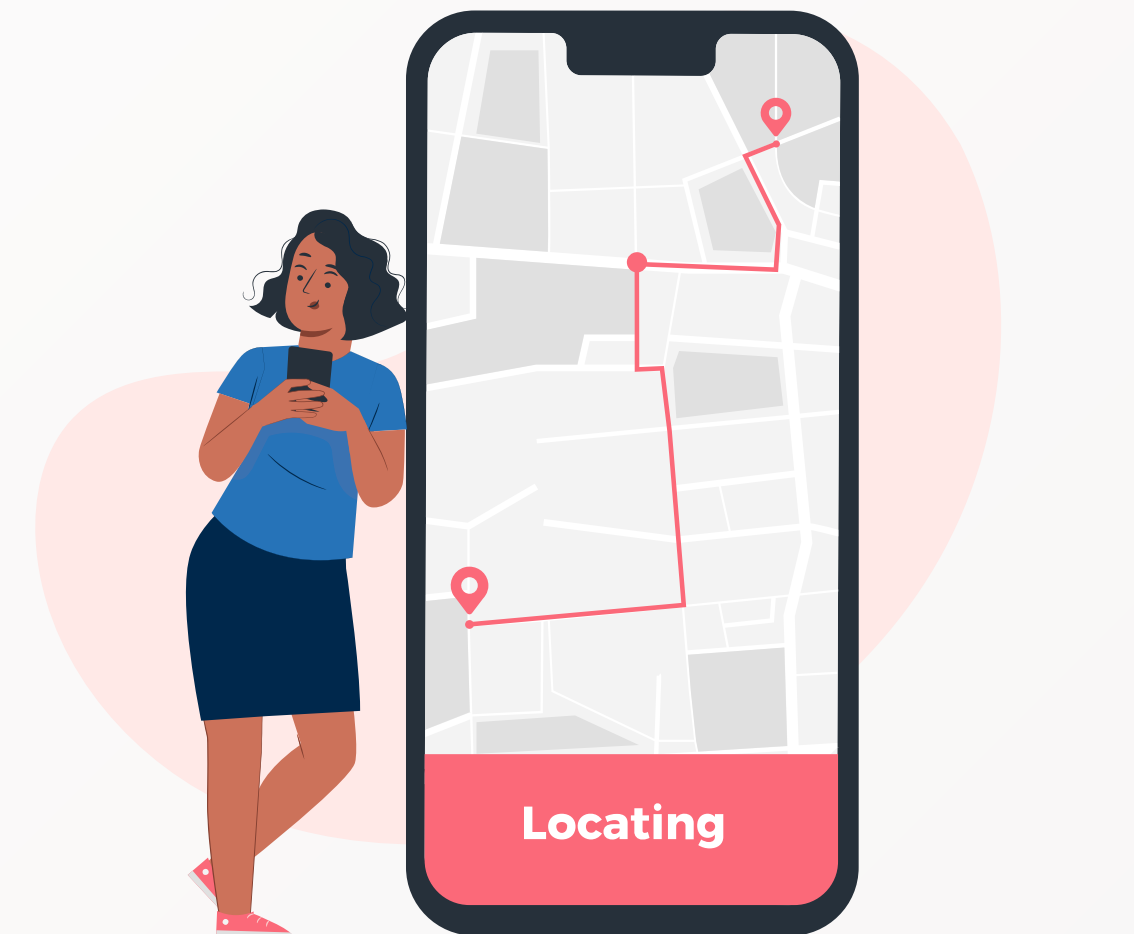
Agile vs. SDLC

Development Aspect	Agile	SDLC
Product Delivery	Rapid through multiple iterations, MVP within 2 weeks	Slow, only after rigorous planning and documentation
Stakeholder Satisfaction	Constant user feedback for iterations	User involvement limited, but surveying conducted
Organisational Fit	Suits students' urgent need for ride-sharing solution	Thorough analysis allows competitive edge against already existing apps

Conclusion

While Agile offers flexibility, SDLC is more effective in comprehensive planning and thorough documentation.

Initial MVP may not hook user interest if it does not set itself apart from existing systems, causing in loss of stakeholder support and distrust.



Thank you!

Welcoming any and all questions

